

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$B_{\mu\nu}$	H	$W_{\mu\nu}$	$H \ell$
T3-d3-d5-F4	0	3	0	5	1	4	$\frac{1}{2}$	$ \begin{aligned} & -\hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) B_{\mu_1 \mu_2}^2 \\ & -\frac{5}{12} \hbar \log \left(\frac{\mu \text{bar}^2}{M_{S_2}^2} \right) B_{\mu_1 \mu_2}^2 \end{aligned} $	$ \begin{aligned} & +C_{H2} H_{a_1} H_{a_1}^\dagger \\ & -H_{a_1} \mu^2 H_{a_1}^\dagger \\ & +\frac{3\text{Iambda}dH \text{IInv}^2 h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{2\epsilon} \\ & +\frac{15\text{Iambda}dH \text{IInv}^2 h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{14\epsilon} \\ & +\frac{95\text{Iambda}dH \text{IInv}^2 h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{42\epsilon} \\ & +\frac{10\sqrt{3}\text{Iambda}dH \text{IInv} \text{Ila mbd}H \text{IInv} 2h H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{7\epsilon} \\ & +\frac{3}{2}\text{Iambda}dH \text{IInv}^2 h \log \left(\frac{\mu \text{bar}^2}{M_{S_1}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\ & +\frac{15}{14}\text{Iambda}dH \text{IInv}^2 h \log \left(\frac{\mu \text{bar}^2}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\ & +\frac{95}{42}\text{Iambda}dH \text{IInv}^2 h \log \left(\frac{\mu \text{bar}^2}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger \\ & +\frac{10\sqrt{2}\text{Iambda}dH \text{IInv} \text{Ila mbd}H \text{IInv} 2h \log \left(\frac{\mu \text{bar}^2}{M_{S_2}^2} \right) H_{a_1} H_{a_2} H_{a_1}^\dagger H_{a_2}^\dagger}{7} \\ & +\frac{2\sqrt{3}\hbar\lambda\sqrt{3}H_{a_1}H_{a_2}\lambda\tau_3H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon} \\ & +\frac{2\sqrt{3}\hbar\lambda\sqrt{3}H_{a_1}H_{a_2}M_{S_1}^2\lambda\tau_3H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\ & +\frac{2\sqrt{3}\hbar \log \left(\frac{\mu \text{bar}^2}{M_{S_1}^2} \right) \lambda\tau_3 H_{a_1} H_{a_2} M_{S_1}^2 \lambda\tau_3 H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\ & -\frac{2\sqrt{3}\hbar\lambda\sqrt{3}H_{a_1}H_{a_2}M_{S_2}^2\lambda\tau_3H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \\ & -\frac{2\sqrt{3}\hbar \log \left(\frac{\mu \text{bar}^2}{M_{S_2}^2} \right) \lambda\tau_3 H_{a_1} H_{a_2} M_{S_2}^2 \lambda\tau_3 H_{a_1}^\dagger H_{a_2}^\dagger}{M_{S_1}^2 - M_{S_2}^2} \end{aligned} $	$ \begin{aligned} & -\frac{6\hbar \left(W_{\mu_1 \mu_2}^A \right)^2}{\epsilon} \\ & -5\hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \left(W_{\mu_1 \mu_2}^A \right)^2 \\ & -\frac{1}{6}\hbar \log \left(\frac{\mu \text{bar}^2}{M_{S_1}^2} \right) \left(W_{\mu_1 \mu_2}^A \right)^2 \\ & -\frac{5}{6}\hbar \log \left(\frac{\mu \text{bar}^2}{M_{S_2}^2} \right) \left(W_{\mu_1 \mu_2}^A \right)^2 \end{aligned} $	$ \begin{aligned} & \frac{2\sqrt{2}\hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \overline{w}_{1,1,1,1} \overline{w}_{2,1,1,1} \ell_{a_1,1,2}^\tau C P_{\ell} \ell_{a_2,1,1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda \tau_1 \ell_{a_1,1,2} \ell_{a_2,1,1} M_F^2}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_1}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \\ & \frac{2\sqrt{2}\hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \overline{w}_{1,1,1,1} \overline{w}_{2,1,1,1} \ell_{a_1,1,2}^\tau C P_{\ell} \ell_{a_2,1,1} H_{a_1} H_{a_2} M_{S_1}^2 \lambda \tau_1 \ell_{a_1,1,2} \ell_{a_2,1,1} M_F^2}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_1}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \\ & \frac{2\sqrt{2}\hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \overline{w}_{1,1,1,1} \overline{w}_{2,1,1,1} \ell_{a_1,1,2}^\tau M_{S_1}^2 \overline{w}_{1,1,1,1} \overline{w}_{2,1,1,1} \ell_{a_1,1,2} \ell_{a_2,1,1} H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_1}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \\ & \frac{2\sqrt{2}\hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \overline{w}_{1,1,1,1} \overline{w}_{2,1,1,1} \ell_{a_1,1,2}^\tau C P_{\ell} \ell_{a_2,1,1} M_{S_1}^2 \overline{w}_{1,1,1,1} \overline{w}_{2,1,1,1} \ell_{a_1,1,2} \ell_{a_2,1,1} H_{a_1}^\dagger H_{a_2}^\dagger M_F^2}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_1}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \\ & \frac{2\sqrt{2}\hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \overline{w}_{1,1,1,1} \overline{w}_{2,1,1,1} \ell_{a_1,1,2}^\tau C P_{\ell} \ell_{a_2,1,1} H_{a_1} H_{a_2} M_{S_1}^2 M_{S_2}^2 \lambda \tau_1 \ell_{a_1,1,2} \ell_{a_2,1,1} M_F}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_1}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \\ & \frac{2\sqrt{2}\hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \overline{w}_{1,1,1,1} \overline{w}_{2,1,1,1} \ell_{a_1,1,2}^\tau C P_{\ell} \ell_{a_2,1,1} H_{a_1} H_{a_2} M_{S_1}^2 M_{S_2}^2 \lambda \tau_1 \ell_{a_1,1,2} \ell_{a_2,1,1} M_F}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_1}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \\ & \frac{2\sqrt{2}\hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \overline{w}_{1,1,1,1} \overline{w}_{2,1,1,1} \ell_{a_1,1,2}^\tau C P_{\ell} \ell_{a_2,1,1} H_{a_1} H_{a_2} M_{S_1}^2 M_{S_2}^2 \lambda \tau_1 \ell_{a_1,1,2} \ell_{a_2,1,1} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_1}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \\ & \frac{2\sqrt{2}\hbar \log \left(\frac{\mu \text{bar}^2}{M_F^2} \right) \overline{w}_{1,1,1,1} \overline{w}_{2,1,1,1} \ell_{a_1,1,2}^\tau C P_{\ell} \ell_{a_2,1,1} M_{S_1}^2 M_{S_2}^2 \lambda \tau_1 \ell_{a_1,1,2} \ell_{a_2,1,1} H_{a_1}^\dagger H_{a_2}^\dagger M_F}{\left(M_F^2 - M_{S_1}^2 \right) \left(M_F^2 - M_{S_1}^2 \right) \left(M_{S_1}^2 - M_{S_2}^2 \right)} \end{aligned} $

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	e
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